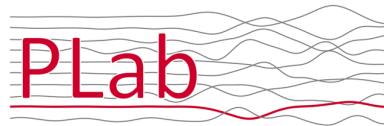


Abstractness avoidance in Maga Rukai

Jennifer Kuo, Cornell University

AMP 2025



Outline

Question: How learnable are abstract URs?

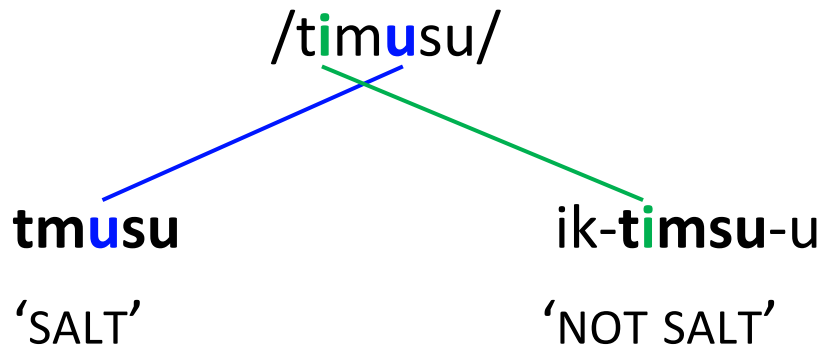
Test case: an alternation pattern in Maga Rukai that requires positing relatively abstract URs

Method: diachronic change as a window into learnability

Finding: Rukai has been restructured in a way that makes it less abstract (and less opaque)

“Typical” morphophonemic analyses allow us to posit abstract URs

An example from Rukai (Hsin 2001)



Resulting Abstractness

No single surface allomorph tells us about all the underlying vowels.

UR is “**composite**”

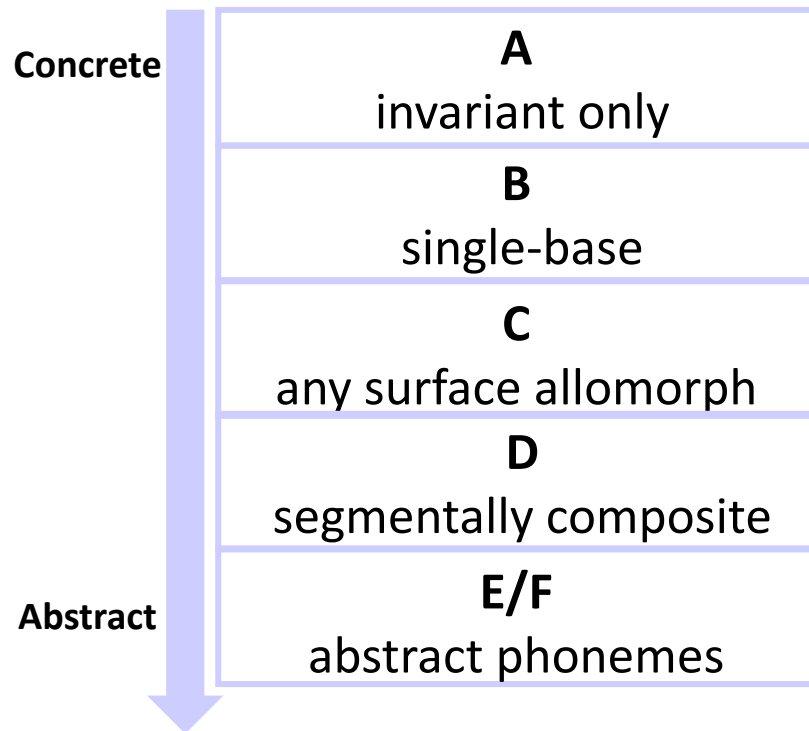
The more abstract URs are allowed to be, the **more difficult UR learning becomes** (larger search space of possible URs)

The abstractness debate

KK-levels:

Kenstowicz & Kisseberth 1977, ch. 1

Wang and Hayes, 2024



The abstractness debate

KK-levels:

Kenstowicz & Kisseberth 1977, ch. 1
Wang and Hayes, 2024

UR search space

[tmsu]~[ik-timsu-u]

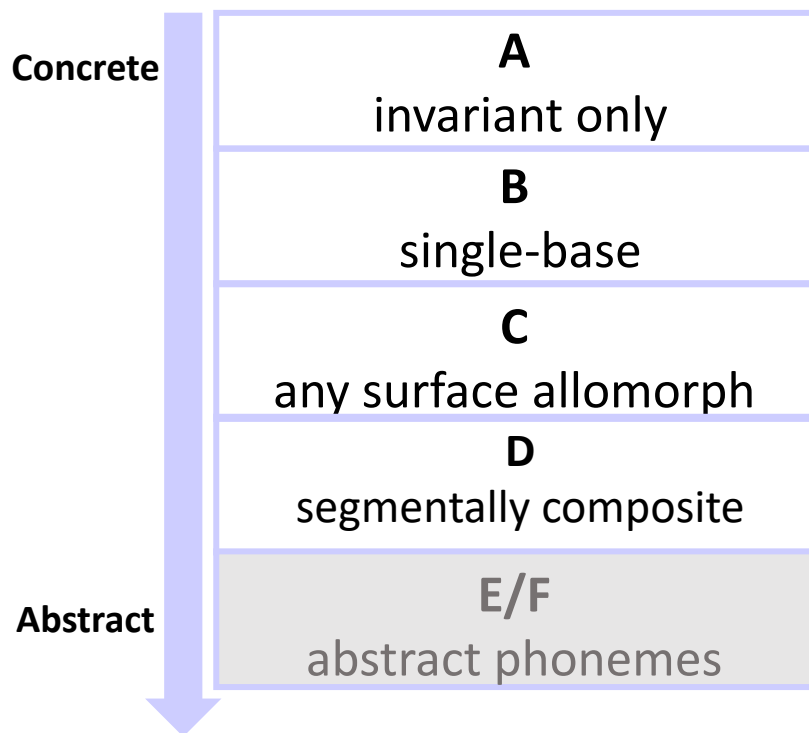
Concrete	A invariant only	/ tmsu /
	B single-base	/ tmsu /
	C any surface allomorph	/ tmsu / , /timsu/
	D segmentally composite	/ timsu / , / tmsu / , /timsu/ , /timus/ , /tims/ , /tmus/...
Abstract	E/F abstract phonemes	

The abstractness debate

KK-levels:

Kenstowicz & Kisseberth 1977, ch. 1

Wang and Hayes, 2024



Support: Albright (2002, 2010, et seq); Dowd (2005)

Related to: listed allomorphs (Zuraw 2000; 2010)

← **Are the KK-D URs of Maga Rukai learned by speakers?**

Evidence against: Kiparsky (1968, 1973, 1982); Derwing (1973); Hooper (1976); Tranel (1981); Gouskova (2012), Rysling (2016), **and more**

Language details – Maga Rukai

Austronesian language spoken
in Southern-Central Taiwan

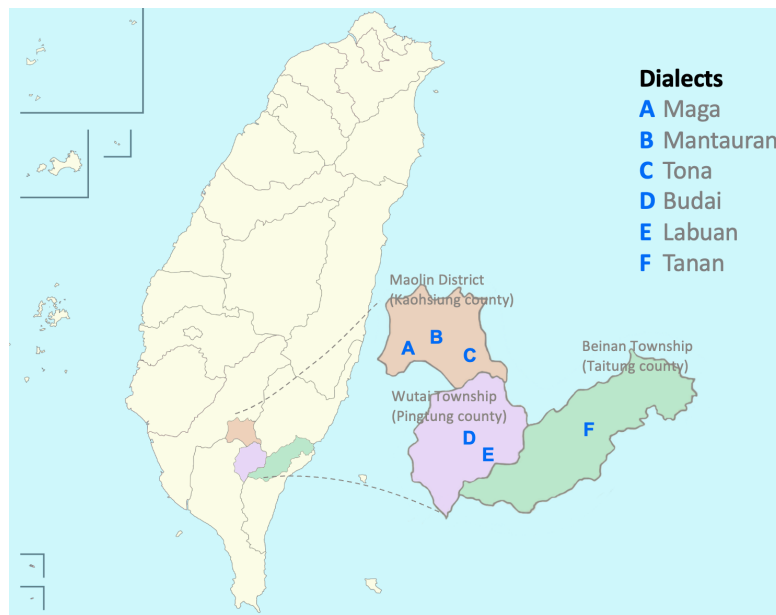
Dialects: **Maga**, Tanan, Budai,
Tona, Labuan, Mantauran

Low mutual intelligibility

Maolin Township

population ~866

Much fewer speakers



Rhythmic syncope: medial vowels are deleted in odd syllables counting from the left edge

Prefixation shifts the position of deleted vowels

Derivation for 'shine'

	STEM	**NEGATIVE, i(k)- -V
UR	/li mi li ma ci/	/i - li mi li ma ci/
	1 2 3 4 5	1 2 3 4 5 6
Syncope	[l_ mi l_ ma ci]	[i li m_ li m_ ci:]
SR	[lmilmaci]	[ilimlimci:]

**Negation is indicated by the prefix /i-(k-) /and lengthening of the final vowel.

Not all affixes trigger vowel alternations

<i>Affix</i>	<i>Gloss</i>	<i>Bare stem</i>	<i>Example</i>	<i>Unattested</i>	
ma-	stative	rkɪmɪ	ma-rkɪmɪ	*ma-rɪkɪmɪ	 no syncope alternations
u-	dynamic vb	θlibi	u-θlibi	*u-θilbi	
si-	to put on	kpiŋi	si-kpiŋi	*si-kipŋi	
i- -V	negation	pliŋɪ	i-pilŋɪ:	*i-pliŋɪ:	
-a:	imperative	skulu	sukl-a:	*skul-a:	 suffix
l-	plural	vlakɪ	l-valkɪ:	*l-vlakɪ:	

Not all affixes trigger vowel alternations

<i>Affix</i>	<i>Gloss</i>	<i>Bare stem</i>	<i>Example</i>	<i>Unattested</i>	
ma-	stative	rkimiɪ	ma-rkimiɪ	*ma-rikmɪ	no syncope alternations
u-	dynamic vb	θlibi	u-θlibi	*u-θilbi	
si-	to put on	kpiŋi	si-kpiŋi	*si-kipŋi	
i- -V	negation	pliŋɪ	i-piŋɪ:	*i-pliŋɪ:	
-a	imperative	skulu	sukl-a:	*skul-a:	focus on this one
l-	plural	vlaki	l-valki:	*l-vlaki:	

Diachronic development of rhythmic syncope

'shine'

'shine.NEG'

1. Alternating stress language

(li, **mi**)(li' **ma**)cɪ

(i, **li**)(mi, **li**)(ma' **cɪ**:)

2. Deletion of unstressed vowels

(l_, **mi**)(l_ ' **ma**)cɪ

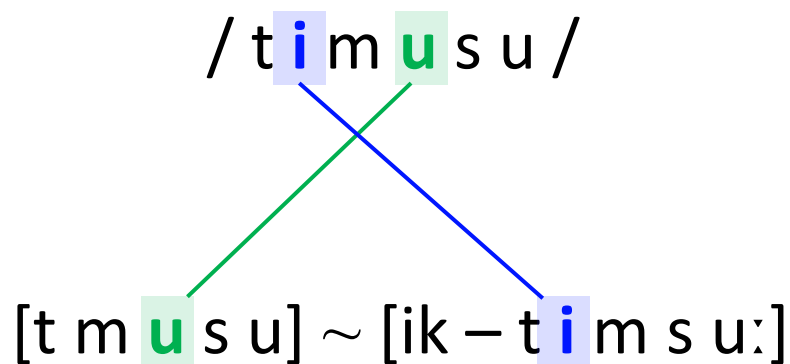
(i, **li**)(m_, **li**)(m_ ' **cɪ**:)

[lmiⁱlma^acɪ]

[i-liⁱmlⁱimcɪ:]

Big picture

Q: How stable is rhythmic syncope in Rukai (and therefore, how learnable are composite URs)?

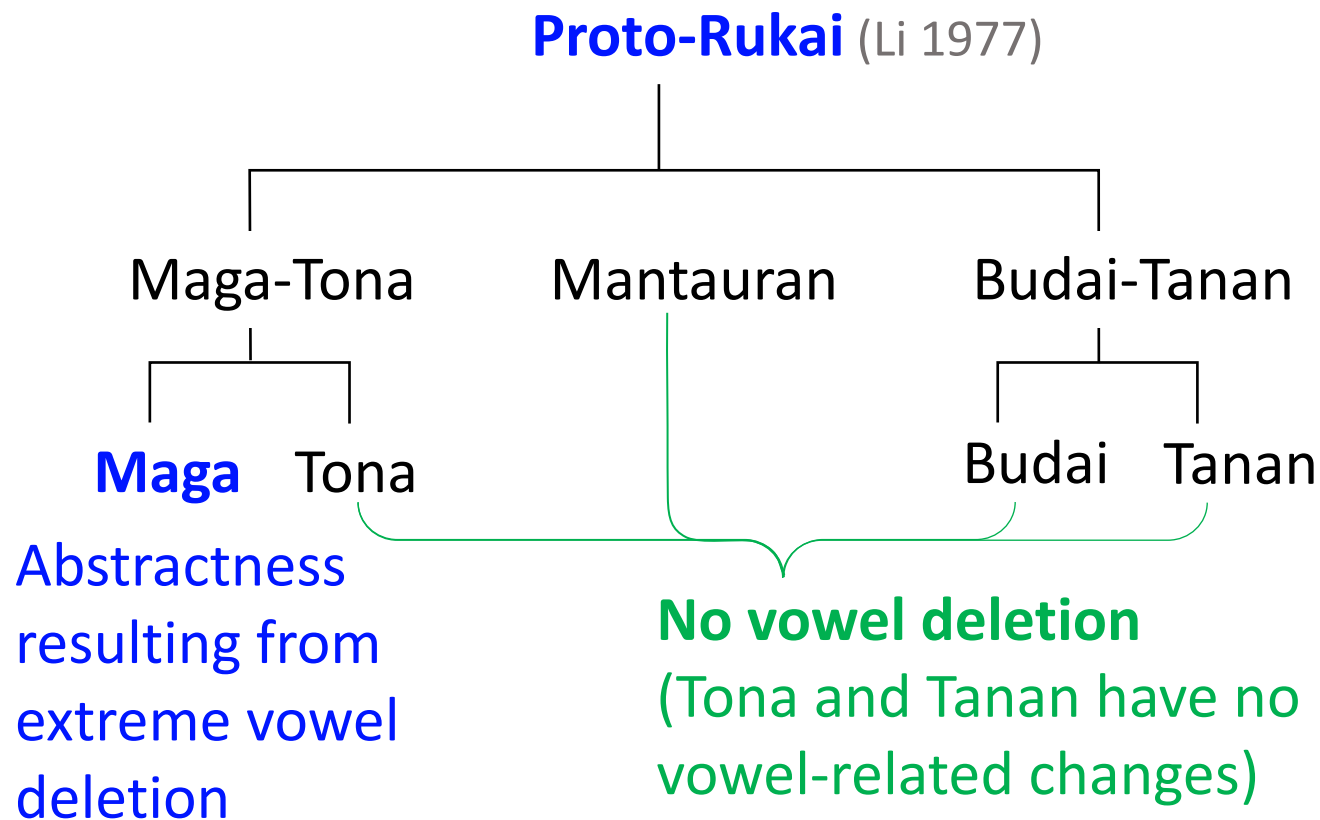


Method: Look at **patterns of paradigm restructuring** in Maga Rukai, by comparing data from Proto-Rukai to Maga-Rukai

~500 Proto-Rukai forms (Li 1977)

959 Maga Rukai paradigms (Hsin 2001)

Proto-Rukai provides a reliable picture of
what vowels looked like before structuring



How abstract does the lexicon actually need to be in Maga?

Not all paradigms require composite URs, because of restrictions on syncope.

Onset-less vowels don't delete [N=194/206, 94%]

<i>UR</i>	<i>SR</i>	<i>gloss</i>
/abu/	<u>a</u> bu (*bu)	'ash'
/alapi/	<u>a</u> lapi (*lapi)	'slate'
/takuasi/	tku <u>a</u> si (cf. *takusi)	'bamboo partridge'

How abstract does the lexicon actually need to be in Maga?

Not all paradigms require composite URs, because of restrictions on syncope.

Vowels in hiatus don't delete [N=23/23, 100%]

<i>UR</i>	<i>SR</i>	<i>gloss</i>
/tanakiasɪ/	tnak <u>i</u> asɪ (*tnakasɪ)	'egret'
/kuluenɪ/	klu <u>e</u> ɪɪ (*klunɪ)	'sparrow'
/i-kucia-V/	i-kic <u>i</u> a: (cf. *i-kica:)	'scissors'

How abstract does the lexicon actually need to be in Maga?

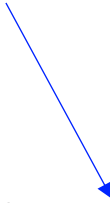
Not all paradigms require composite URs, because of restrictions on syncope.

PREDICTABLE FROM...	UR	STEM	NEGATIVE	
both allomorphs	/abu/	abu	i-abu:	deleting position, but initial/onsetless
stem allomorph	/abaki/	abaki	ik-abki:	
negative allomorph	/rana/	rna:	ik-rana:	
neither (composite)	/timusu/	tmusu	ik-timsu:	not in a deleting position

How abstract does the lexicon actually need to be in Maga?

Not all paradigms require composite URs, because of restrictions on syncope.

In a deleting position,
but is initial/onsetless



PREDICTABLE FROM...	UR	STEM	NEGATIVE
both allomorphs	/abu/	abu	i-abu:
stem allomorph	/abaki/	abaki	ik-abki:
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negative allomorph	/rana/	rna:	ik-rana:
neither (composite)	/timusu/	tmusu	ik-timsu:

no vowels
in deleting
position



How abstract does the lexicon actually need to be in Maga?

Not all paradigms require composite URs, because of restrictions on syncope.

PREDICTABLE FROM...	UR	STEM	NEGATIVE
both allomorphs	/abu/	abu	i-abu:
stem allomorph	/abaki/	abaki	ik-abki:
negative allomorph	/rana/	rna:	ik-rana:
neither (composite)	/timusu/	tmusu	ik-timsu:

← actually composite

Most of the lexicon requires composite URs

Not all paradigms require composite URs, because of restrictions on syncope.

Counts by paradigm type (if syncope applies regularly):

PREDICTABLE FROM...	KK-LEVEL REQUIRED	N	P
both allomorphs	A: invariant	84	0.08
stem allomorph	B/C: surface allomorph	96	0.09
negative allomorph	B/C: surface allomorph	230	0.22
neither	D: composite	639	0.61

most of the lexicon



Proposal (and preview of results)

Maga rhythmic syncope has been restructured as **metathesis and vowel-copying** (that is exceptionally triggered by some morphemes)
making the pattern **more concrete**

Observation: alternating vowels
tend to match each other

Matching

[θl <u>i</u> bi] ~ [i-θl <u>i</u> bi:]	‘fly’
[d <u>m</u> arɪ] ~ [i-k-d <u>a</u> mri:]	‘moon’
[rn <u>a</u> :] ~ [ik-r <u>a</u> na:]	‘creek’
[ab <u>a</u> kɪ] ~ [ik- <u>a</u> bki:]	‘souls’

Non-matching

[tm <u>u</u> su] ~ [i-k-t <u>i</u> msu:]	‘salt’
[<u>a</u> m <u>i</u> cu] ~ [i- <u>a</u> mcu:]	‘nightshade’
[tbi:] ~ [i-t <u>u</u> bi:]	‘weep’

In Hsin’s corpus, **52%**
(N= 410/790) of
paradigms are
vowel-matching
cf. chance rate = 16.7%

Vowel-matching makes paradigms predictable from both surface allomorphs

Matching

[θl <u>i</u> bi] ~ [i-θ <u>l</u> bi:]	‘fly’
[d <u>m</u> arɪ] ~ [i-k-d <u>a</u> mri:]	‘moon’
[rn <u>a</u> :] ~ [ik-r <u>a</u> na:]	‘creek’
[ab <u>a</u> kɪ] ~ [ik- <u>a</u> bki:]	‘souls’

Non-matching

[tmusu] ~ [i-k-timsu:]	‘salt’
[<u>a</u> m <u>i</u> cu] ~ [i- <u>a</u> mcu:]	‘nightshade’
[tbi:] ~ [i-t <u>u</u> bi:]	‘weep’

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[rn <u>a</u> :] ~ [ik-r <u>a</u> na:]	‘creek’
[ab <u>a</u> kɪ] ~ [ik- <u>a</u> bki:]	‘souls’

Looks like metathesis

stem → *negative*

UR (=stem) /i-k-dm₁a₂ri/

SR [i-k-da₂m₁ri:]

Non-matching

[tmusu] ~ [i-k-timsu:]	‘salt’
udalɪ i-udalɪ	‘rain’
u-tbi: ~ i-tubi:	

Vowel-matching makes paradigms predictable from both surface allomorphs

Matching

[θl <u>i</u> bi] ~ [i-θ <u>i</u> lbi:]	‘fly’
[d <u>m</u> arɪ] ~ [i-k-d <u>a</u> mri:]	‘moon’
[rn <u>a</u> :] ~ [ik-r <u>a</u> na:]	‘creek’
[ab <u>a</u> kɪ] ~ [ik- <u>a</u> bki:]	‘souls’

Looks like vowel copying

stem → *negative*

UR (=stem) /i-k-rna₁-V/
SR [i-k-ra₁na₁:]



Non-matching

[tmusu] ~ [i-k-timsu:]	‘salt’
udalɪ i-udalɪ	‘rain’
u-tbi ~ i-tubi:	

Similar patterns of
predictability across
all paradigm types

Prediction:

If learners avoid abstract URs, then vowel-related changes should result in **more vowel-matching**

Proto-Rukai	Expected SRs	Actual SRs	Gloss
* <u>s</u> amito	smitu ~ i-k-s <u>a</u> mtu:	smitu ~ i-k-s <u>i</u> mtu:	'lips'
*t <u>o</u> bakə	tbaki ~ i-k-t <u>u</u> bki:	tbaki ~ i-k-t <u>a</u> bki:	'shell'

Result:

Vowel-related changes between Proto-Rukai and Maga Rukai have resulted in **more vowel-matching**

Proto-Rukai	Expected SRs	Actual SRs	Gloss
* <u>s</u> amito	smitu ~ i-k-s <u>a</u> mtu:	smitu ~ i-k-s <u>i</u> mtu:	‘lips’
* <u>t</u> obakə	tbaki ~ i-k-t <u>u</u> bki:	tbaki ~ i-k-t <u>a</u> bki:	‘shell’

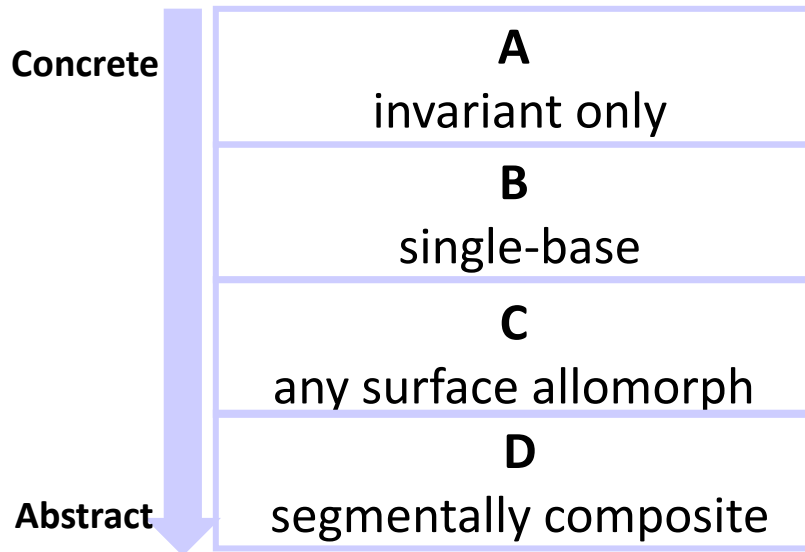
CHANGE RESULTS IN...	N	P	EXAMPLE
MORE vowel matching	39	0.70	*s <u>a</u> m <u>i</u> to → /s <u>i</u> m <u>i</u> tu/
LESS vowel matching	5*	0.09	*v <u>i</u> l <u>i</u> → /v <u>u</u> l <u>i</u> /
neither	12	0.21	*p <u>i</u> l <u>a</u> y → /p <u>u</u> l <u>e</u> :/

*2 explainable through other alternations

Question:

Exactly **what level of abstractness** is permitted in
Maga Rukai?

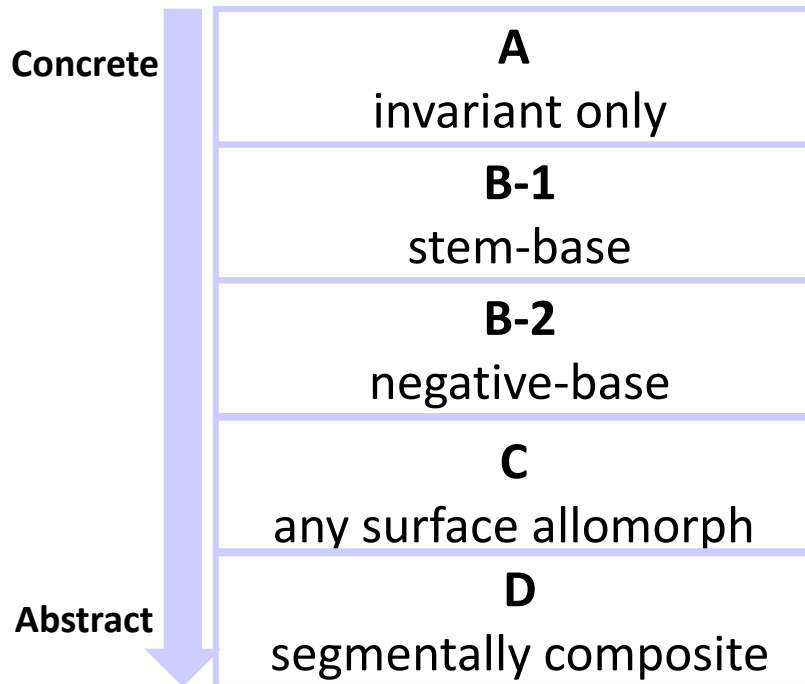
KK-levels:



Question:

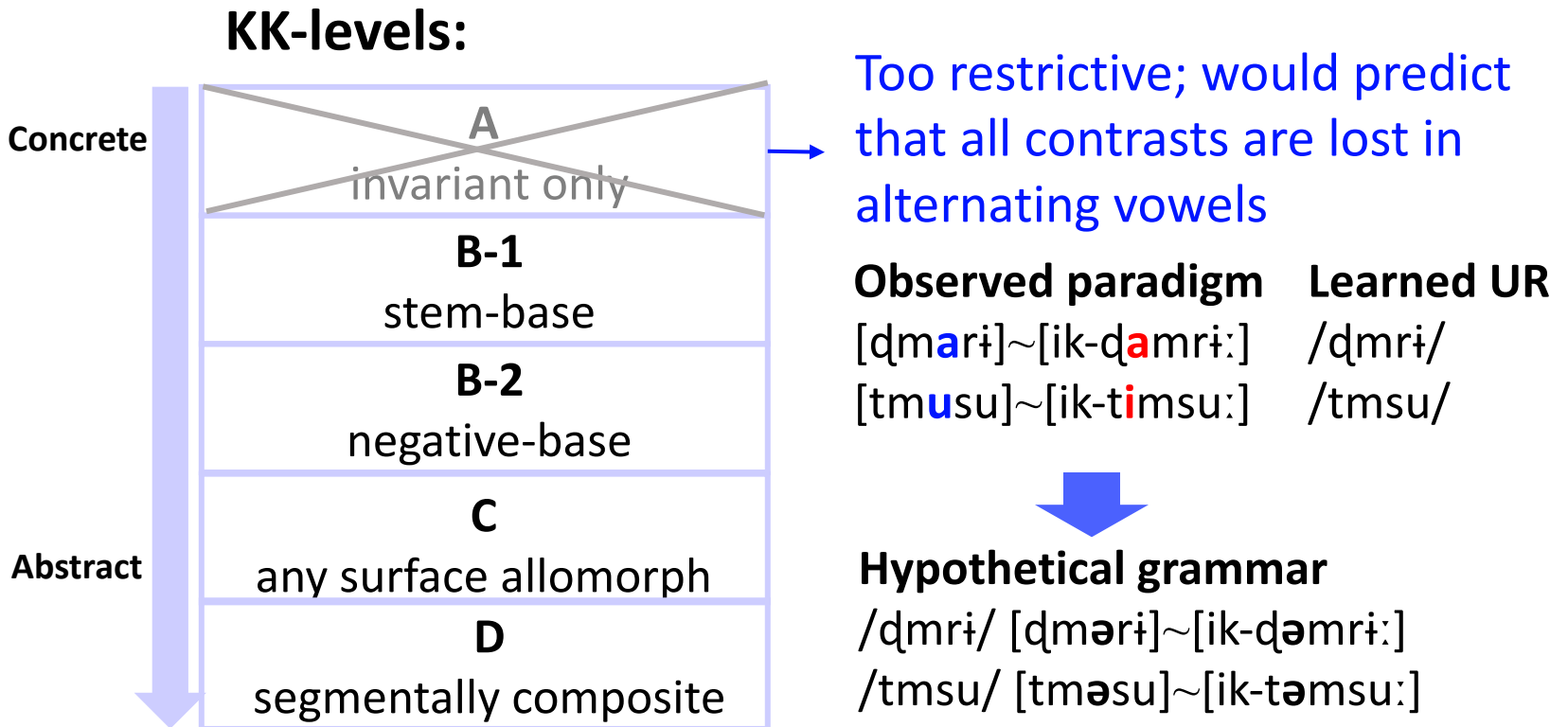
Exactly **what level of abstractness** is permitted in
Maga Rukai?

KK-levels:



Question:

Exactly **what level of the abstractness** is permitted in Maga Rukai?



Question:

Under each level of the KK-hierarchy, which paradigms should undergo vowel-matching changes?

B-1: stem-base

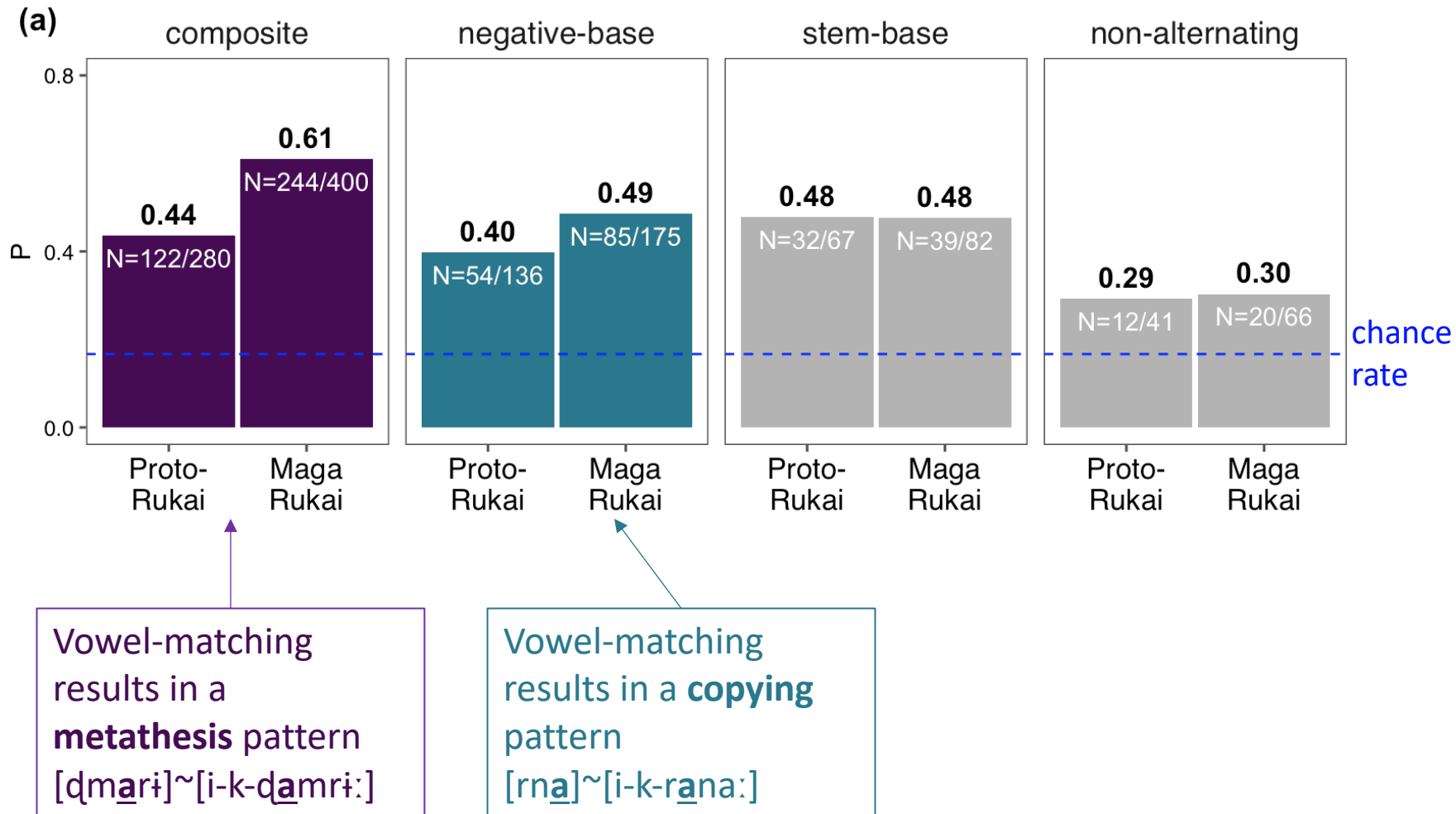
B-2: negative-base

C: any surface allomorph

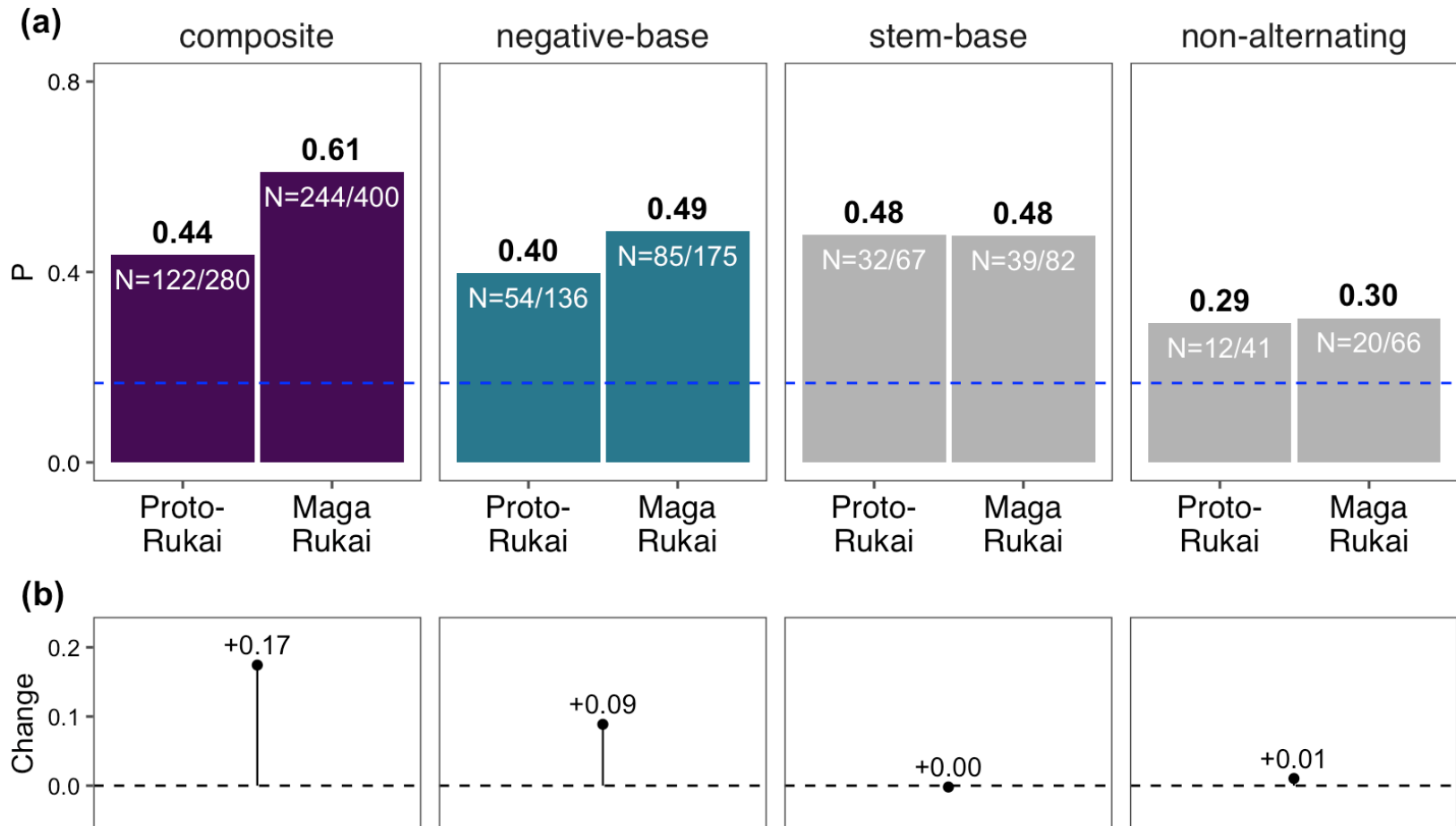
D: segmentally composite

TYPE	UR	STEM	NEGATIVE	B1	B2	C	D
non-alternating	/iri/	i ri	i- i ri:	x	x	x	x
stem-base	/abaki/	a baki	ik- a bki:	x	✓	x	x
negative-base	/rana/	r a :	ik-r a na:	✓	x	x	x
composite UR	/damari/	q a ri	ik-q a mri:	✓	✓	✓	x

Results suggest that vowel-matching increases a paradigm's **predictability from the stem form**



Results suggest that vowel-matching increases a paradigm's **predictability from the stem form**



Question:

Under each level of the KK-hierarchy, which paradigms should undergo vowel-matching changes?

B-1: stem-base

B-2: negative-base

C: any surface allomorph

D: segmentally composite

TYPE	UR	STEM	NEGATIVE	B1	B2	C	D
non-alternating	/iri/	i ri	i- i ri:	x	x	x	x
stem-base	/abaki/	a baki	ik- a bki:	x	✓	x	x
negative-base	/rana/	r a :	ik-r a a:	✓	x	x	x
composite UR	/damari/	q a ri	ik-q a mri:	✓	✓	✓	x

Answer: likely to be KK-B; learners have a strong preference for **stem-based restructuring**

Results are supported by the
direction of vowel-matching changes

Restructuring towards stem

Proto-Rukai:	*s a m i to
Expected paradigm:	sm i tu ~ ik-s a mtu:
Actual paradigm:	sm i tu ~ ik-s i mtu:



Restructuring towards negative

Proto-Rukai:	*s i n a ws i n a w
Expected paradigm:	sn o sno ~ i-s i ns i no:
Actual paradigm:	sn i sno ~ i-s i ns i no:



Vowel-matching changes have been **mostly towards the stem form**

Restructuring towards stem (48/56 = 87%)

Proto-Rukai: ***s****a**m**i**to

Expected paradigm: sm**i**tu ~ ik-s**a**mtu:

Actual paradigm: sm**i**tu ~ ik-s**i**mtu:


Restructuring towards negative (2/56 = 4%) ←

Proto-Rukai: ***s****i**n**a**w**s****i**n**a**w

Expected paradigm: u-s**n**o**s**no ~ i-s**i**n**s****i**no:

Actual paradigm: u-s**n****i**sno ~ i-s**i**n**s****i**no:

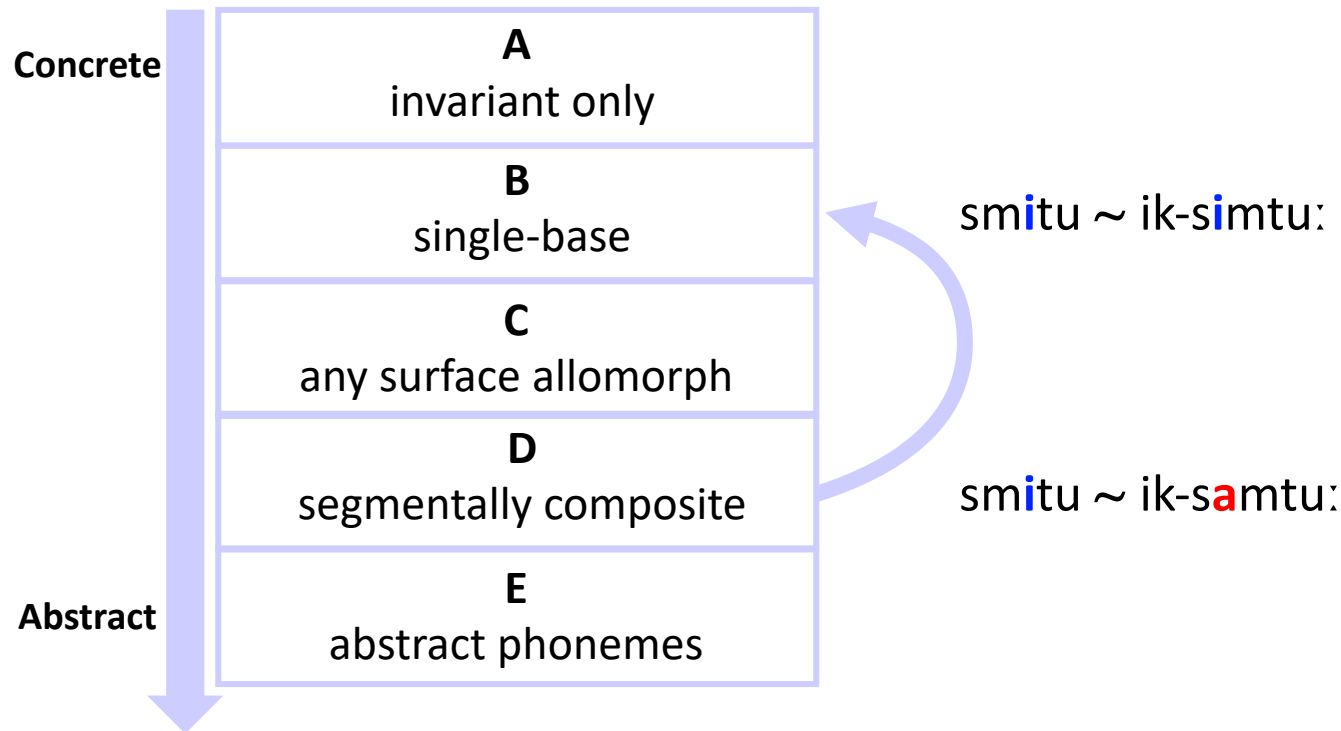

These have other explanations, e.g. CiCo reduplication (Zeitoun 2007; Zeitoun 2024)

Unclear, both Vs changed (5/56, 9%)

Results: speakers are learning URs in line with **level B(1)** of the abstractness hierarchy

KK-levels:

Kenstowicz & Kisseberth 1977, ch. 1
Wang and Hayes, 2024



Caveat: in addition to vowel-matching, paradigms also undergo leveling

In Hsin's corpus of Maga forms, not all paradigms show the expected vowel alternation pattern.

Regularly alternating

tpulu~i-taplu:

amudu ~ ik-amdu:

Exceptions to regular syncope pattern

udali~i-udali: (cf. udali~i-udli:) ← leveled to stem

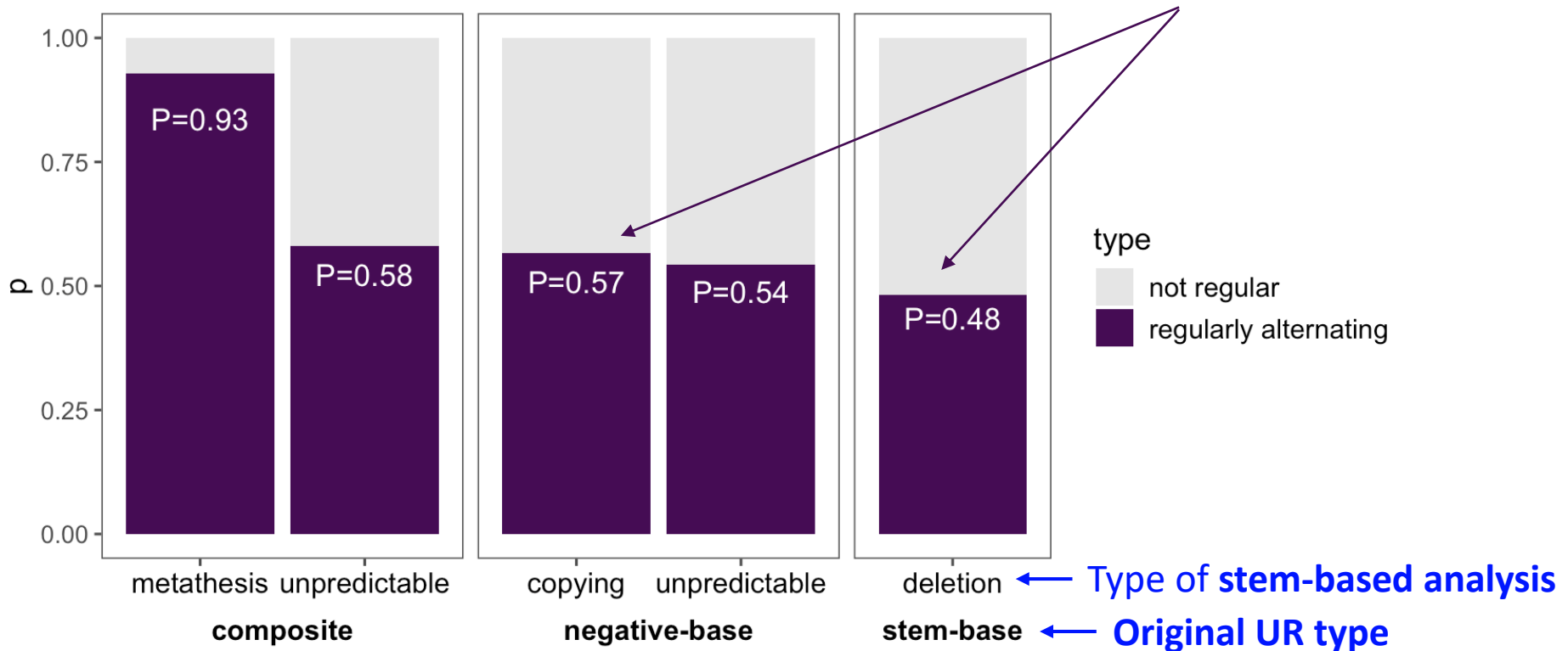
blavni ~ ik-bØlavi: (cf. blavni~ik-bulvani:) ← Leveled to stem

mici ~ i-mici: (cf. mci: ~ i-mici:) ← Leveled to negative

Alternations are **more likely to leveled** for paradigms that **do not look like metathesis**

Figure: rate of regularly alternating forms

*excluding paradigms that were non-alternating to begin with, e.g. [abu] ~ [i-abu]



Patterns of leveling show that...

Leveling can be **towards both allomorphs**, in line with KK-C (choosing from allomorphs) of the abstractness hierarchy.

Vowel-matching increased both metathesis-type and copying-type patterns...

Why do the metathesis-type paradigms get leveled less than the copying-type paradigms?

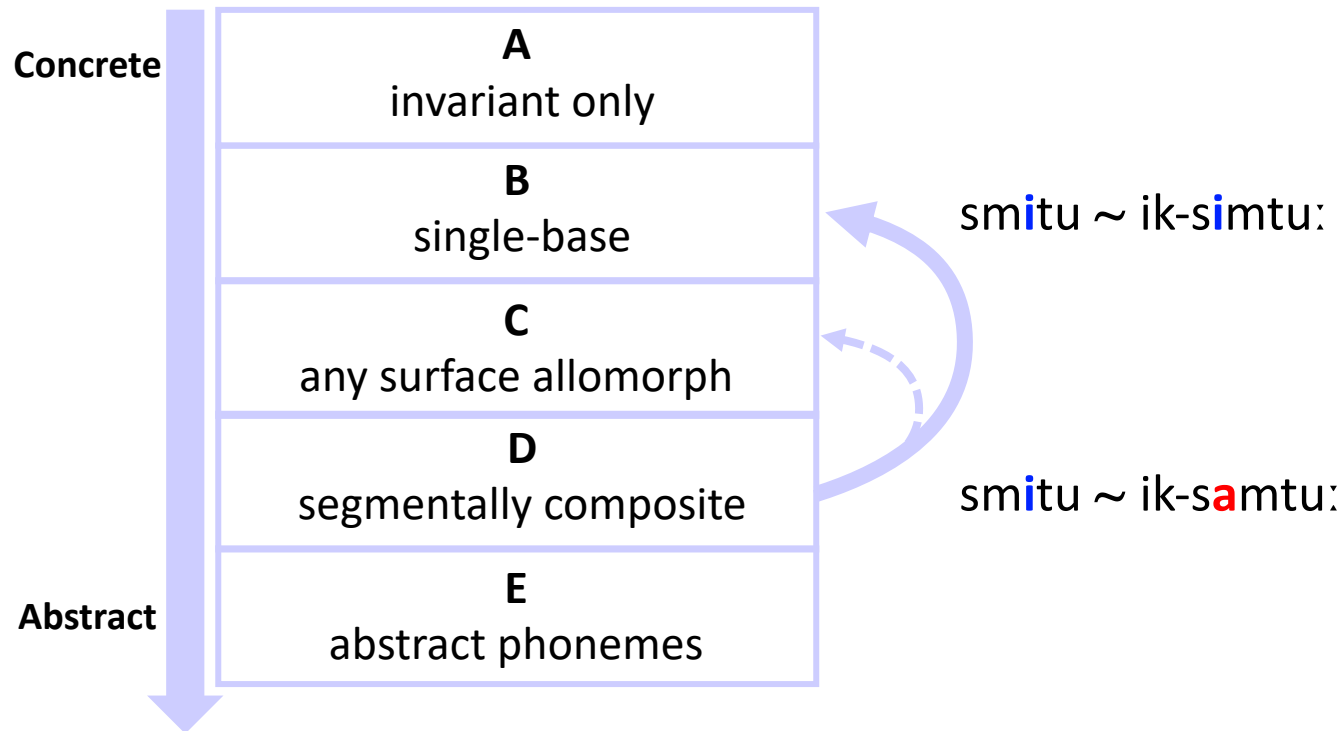
- these are the **most frequent** paradigm type
- leveling is motivated by other **markedness reasons**

Revised results:

speakers **prefer to learn stem-based URs**, but can also learn negative-base URs, in line with **KK-C**

KK-levels:

Kenstowicz & Kisseberth 1977, ch. 1
Wang and Hayes, 2024



Summing up

Result of restructuring: In Maga, an abstract-amenable rhythmic syncope alternation was reanalyzed as a **concrete pattern**.

Direction of restructuring. Has been asymmetrically towards the stem allomorph, providing partial support for **a single-base view of morphophonemic learning**.

The resulting analysis...

- **preserved vowel alternations** to some degree
- is **imperfect**: has exceptions, and is not motivated by global phonological principles

Suggesting that speakers prefer to learn learn an **imperfect but more concrete pattern**, than one that is abstract.

Thank you! takdɨnesu !

Thanks to: members of Cornell phonetics lab, Bruce Hayes, Dustin Bowers, Elizabeth Zeitoun, Abigail Cohn, Maria Gouskova, for their insight and comments 😊

The Rukai alternation pattern...

is **morpheme-specific**

[kpiŋi] ~ [si-kpiŋi] (instrumental)

triggered by both **prefixes and suffixes**

[skulu] ~ [sukl-a:] (imperative)

[ɖmaɾɨ] ~ [i-k-ɖamɾɨ:]

[vlakɨ] ~ [l-valkɨ] (plural, not productive)

Not easily described by **suffixal alignment constraints**

[ɖmaɾɨ ~ ik-ɖamr-ɨ:]

[rkɨɾkɨ ~ ik-rɨkɨk-ɨ:]

One proposal: root structure constraints?

*VCV_{unaffixed}

[**rn**a:]

[**d**maɪ]

[**rk**ɪrki]

[**sk**ulu]

[**vl**aki]

*VCC_{affixed}

[ik-**ra**na:]

[i-k-**d**amri:]

[i-k-**r**ikriki:]

[**su**kl-a:]

[l-**va**lki]

Similar to:

- source vs. product oriented generalizations (Becker & Gouskova 2016)
- Constraints indexed to roots vs. derived stems (Jurgec and Bjorkman 2018)

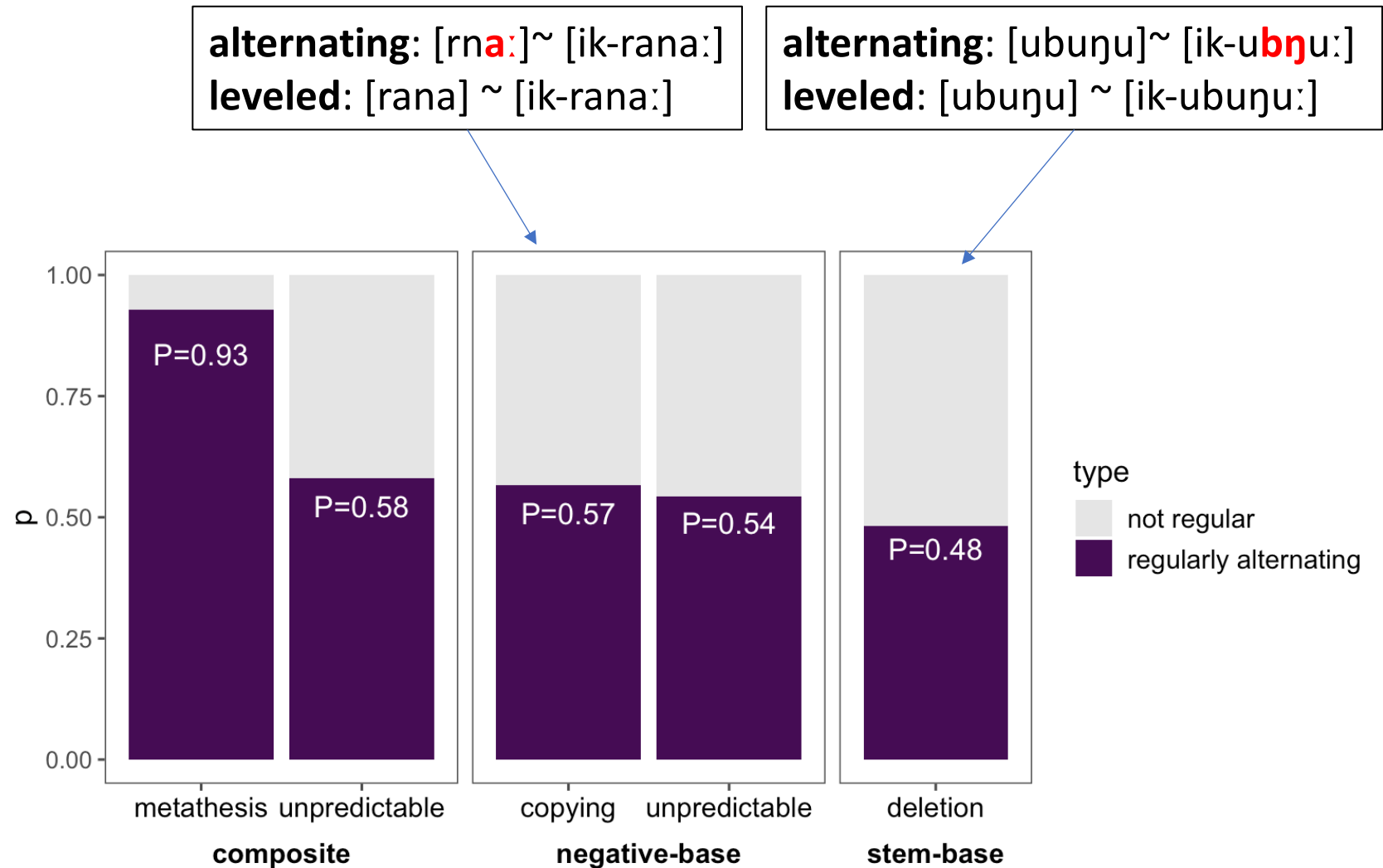
Not easily described by suffixal alignment constraints

[ɖmarɪ̯ ~ ik-ɖ**amr**-ɪ̯:]

[rkɪrkɪ̯ ~ ik-rɪk**ɪk**-ɪ̯:]

Shape of the right edge of the stem changes depending on polarity (even/odd)

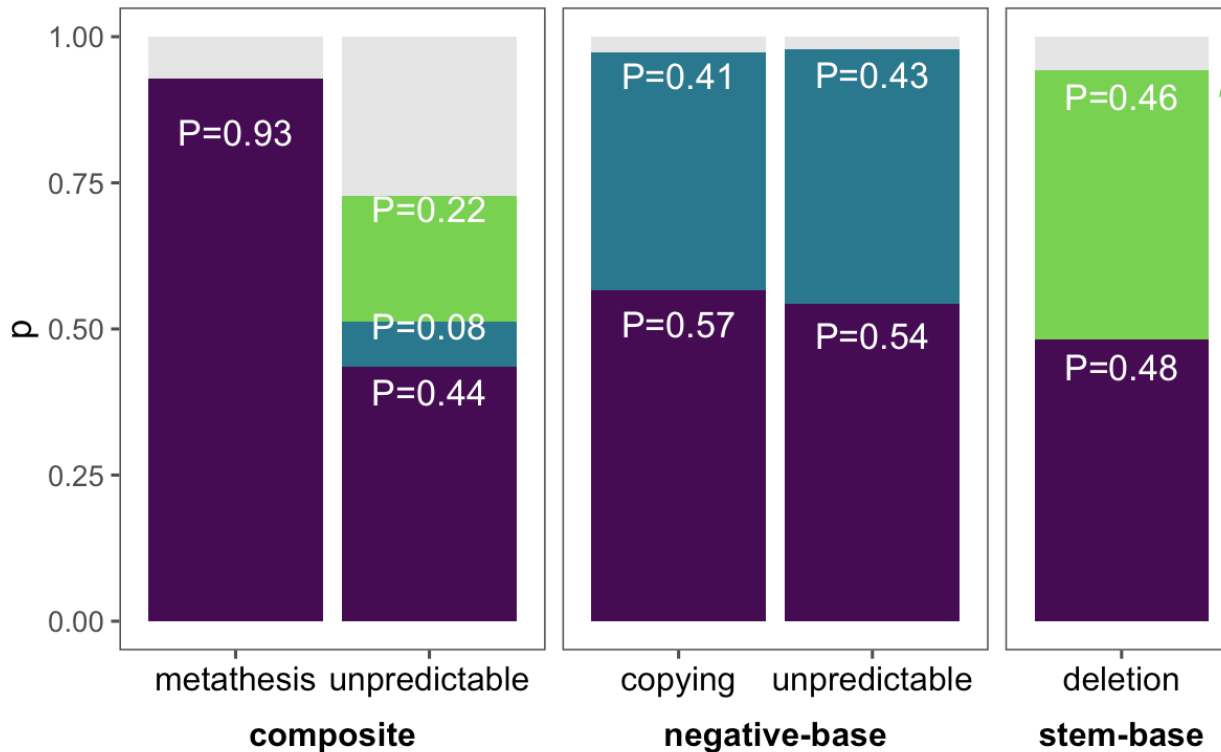
Markedness-driven leveling?



Alternations are **more likely to leveled** for paradigms that **do not look like metathesis**

Figure: rate of alternating forms

*excluding paradigms that were non-alternating to begin with, e.g. [abu] ~ [i-abu]



Stem-base paradigms have **leveled towards the stem**.

E.g. [tru] ~ [ik-turu:]



[turu] ~ [ik-turu:]

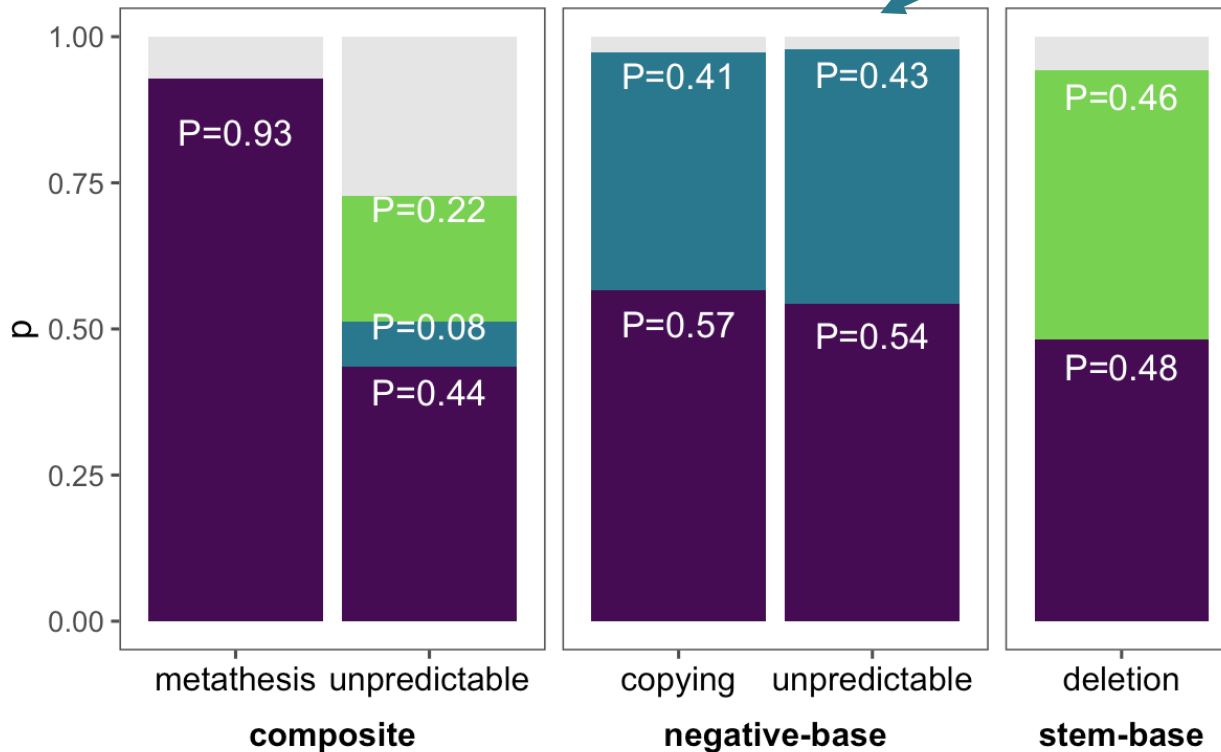
type

- other
- leveled to stem
- leveled to negative
- regularly alternating

Alternations are **more likely to leveled** for paradigms that **do not look like metathesis**

Figure: rate of alternating forms

*excluding paradigms that were non-alternating to begin with, e.g. [abu] ~ [i-abu]



Negative-base paradigms have mostly **leveled towards the negative**.

E.g. [ubali:] ~ [ik-ubli:]

↓
[ubali:] ~ [ik-ubali:]

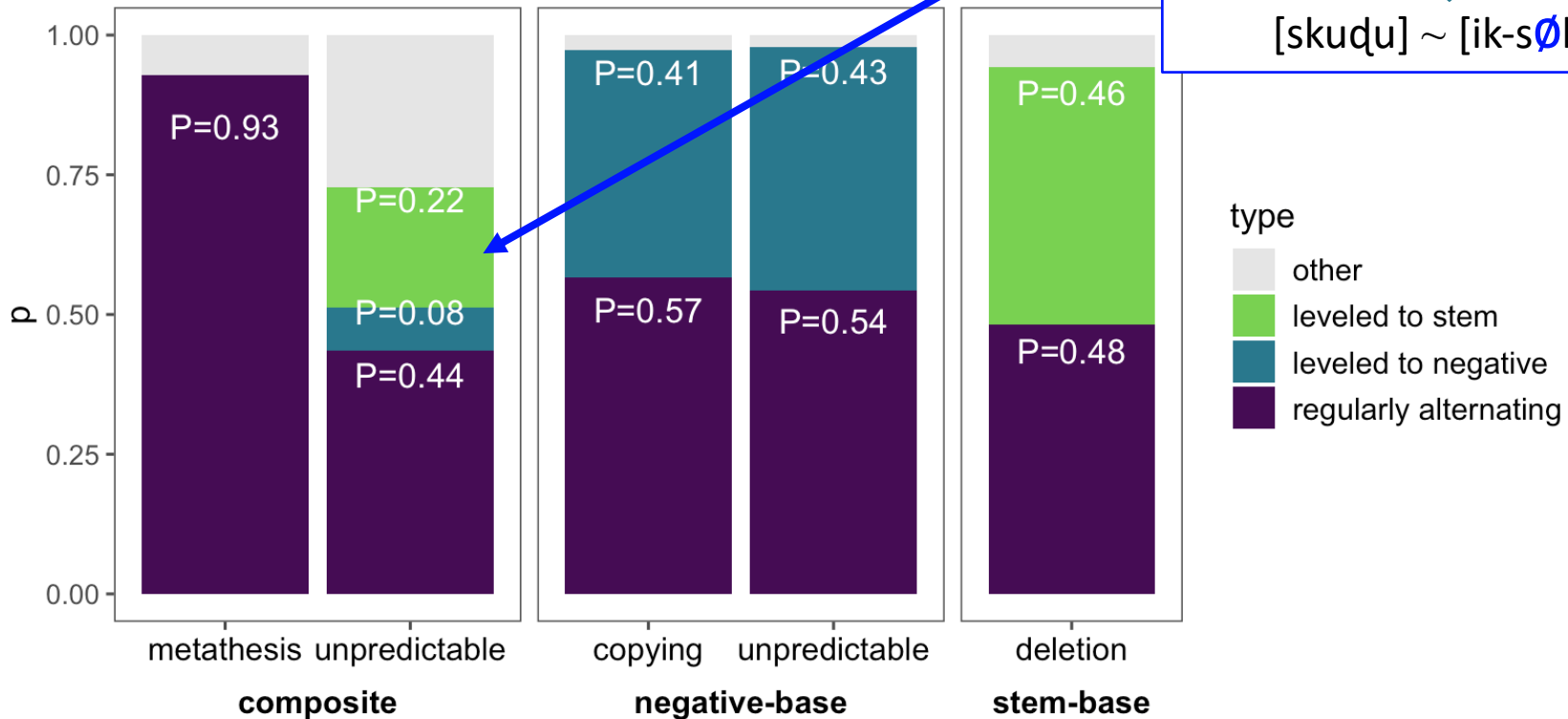
type

- other
- leveled to stem
- leveled to negative
- regularly alternating

Alternations are **more likely to leveled** for paradigms that **do not look like metathesis**

Figure: rate of alternating forms

*excluding paradigms that were non-alternating to begin with, e.g. [abu] ~ [i-abu]



Otherwise, preference for **leveling towards the stem**

E.g. [skuɖu] ~ ik-sakɖu:]



[skuɖu] ~ [ik-sØkuɖu:]

Extra slides...

Remaining problem: **what motivates metathesis?**

Under our new metathesis analysis, 'see' would have UR/SRs:

/cɨɪli/ [cɨɪli] ~ [i-cɪɪli:]

UR is identical to stem allomorph

because evidence suggests that **URs are based on the stem allomorph**

Remaining problem: what motivates metathesis?

Deriving 'see' /cɨɪli/ [cɨɪli] ~ [i-cɨɪli]

Markedness constraint that motivates metathesis

Faithfulness constraint that penalizes metathesis

Faithful candidate
Winner

In the stem, the faithful candidate wins because metathesis is not motivated

/i-cɨɪli/	M	F
a. i-cɨɪli	*!	
→ b. i-cɨɪli		*
/cɨɪli/		
→ c. cɨɪli		
d. cɨɪli		*!

Remaining problem: what motivates metathesis?

Deriving 'see' /cɲɪli/ [cɲɪli] ~ [i-cɪŋli]

Markedness constraint that motivates metathesis

Faithfulness constraint that penalizes metathesis

Wrong winner

/i-cɲɪli/	*CC	LINEARITY
☹ a. i-cɲɪli	*	
b. i-cɪŋli	*	*!
/cɲɪli/		
→ c. cɲɪli	*	
d. cɪŋli	*	*!

Problem: No clear markedness motivation

CC clusters (and complex onsets) are found elsewhere in Maga

Examples: [cɲɪli], [tɲupru], [i-cɪŋli:]

(Tentative) proposal

Metathesis in Maga is driven by **morpheme-specific requirements** rather than the global phonology

Aligns with the typology of metathesis

- Restricted to particular **segments or positions** (Hume & Seo 2004; Sohn 1980)
- Not true transposition, but rather **gestural overlap** (Gilbert & Mooney 2021; Mooney 2022)
- restricted to **particular morphemes** (Butskhrikidze & van de Weijer 2003)

Implementation: lexically indexed constraints

Constraints can be indexed to a morpheme or a set of morphemes (Pater 2006, 2008)

Alternative: co-phonologies (Inkelas and Zoll 2007)

*_L[CC : Assign a violation mark to any CC at the boundary of a morpheme specified as L

Intuition: Metathesis to break up CC clusters is triggered by some prefixes

Approach 1: indexed constraints (Pater 2006, 2008)

Deriving 'see' /cɨŋli/ [cɨŋli] ~ [i-cɨŋli]

Markedness constraint that motivates metathesis

Faithfulness constraint that penalizes metathesis

Faithful candidate
Winner

/i _L -cɨŋli/	* _L [CC]	LINEARITY	*CC
a. i-cɨŋli	*!		*
→ b. i-cɨŋli		*	*
/cɨŋli/			
→ c. cɨŋli			*
d. cɨŋli		*!	*

General markedness constraint is ranked low

Un-prefixed stem does not violate the indexed constraint

Problem: the locus of the constraint violation should be on the prefix (indexed morpheme), but is not.

Approach 2: Syllable alignment

Deriving 'see' /cɨɪli/ [cɨɪli] ~ [i-cɨɪli]

Markedness constraint that motivates metathesis

Faithfulness constraint that penalizes metathesis

/i _L -cɨɪli/	* _L [CC]	LINEARITY	*CC
a. (i-c.ɨɪ).li	*!		*
→ b. (i-cɨɪ).li		*	*
/cɨɪli/			
→ c. cɨɪli			*
d. cɨɪli		*!	*

General markedness constraint is ranked low

Problem: the locus of the constraint violation should be on the prefix (indexed morpheme), but is not.

Finishing up the analysis - longer words

Deriving 'shrimp' /tkasluɖu/ [tkasluɖu] ~ [i-t**aksul**ɖu]

*CCC forces metathesis of other vowels to fit general well-formedness constraints

Faithful
metathesizes V1
Winner

/i _L -tkasluɖu/	* _L [CC	* CCC	LINEARITY	*CC
a. i-tkasluɖu	*!			**
b. i-ta k sluɖu		*!	*	**
→ c. i-taksulɖu			**	**

{ *_L[CC , MAX-C, *CCC } >> LINEARITY >> *CC

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(Metrical) opacity

Syncope must happen after building the metrical structure which conditions it; put another way, syncope removes the environment that conditions it

	UR	/timusu/	/tavanulu/
Metrical Struc.		(ti. mu).su	(ta. va)(ηulu)
Syncope		tmu.su	tvaŋ. lu
Assign stress		tmú.su	tván. lu
SR		[tmúsu]	[tvánlu]

Interaction of syncope and stress assignment by syllable parity

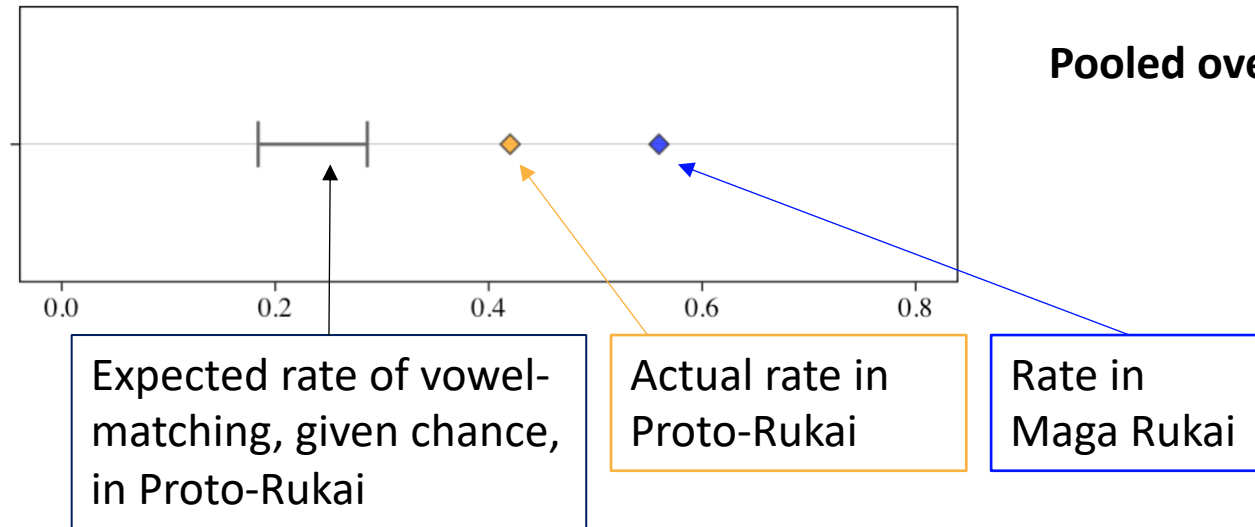
Parity	UR	SR	Example		
Even	/σ ₁ σ ₂ /	[∅ ₁ 'σ ₂]	/t í ba/	tbá:	‘wild yam’
	/σ ₁ σ ₂ σ₃ σ ₄ /	[∅ ₁ 'σ ₂ ∅ ₃ σ ₄]	/tavan u lu/	tvánlu	‘chili pepper’
Odd	/σ ₁ σ₂ σ ₃ /	[∅ ₁ 'σ ₂ σ ₃]	/ɖam a ri/	ɖmári	‘moon’
	/σ ₁ σ ₂ σ ₃ σ₄ σ ₅ /	[∅ ₁ σ ₂ ∅ ₃ 'σ ₄ σ ₅]	/takasul u ɖu/	tkaslúɖu	‘shrimp’

Restructuring that removes opacity, but not abstractness of URs

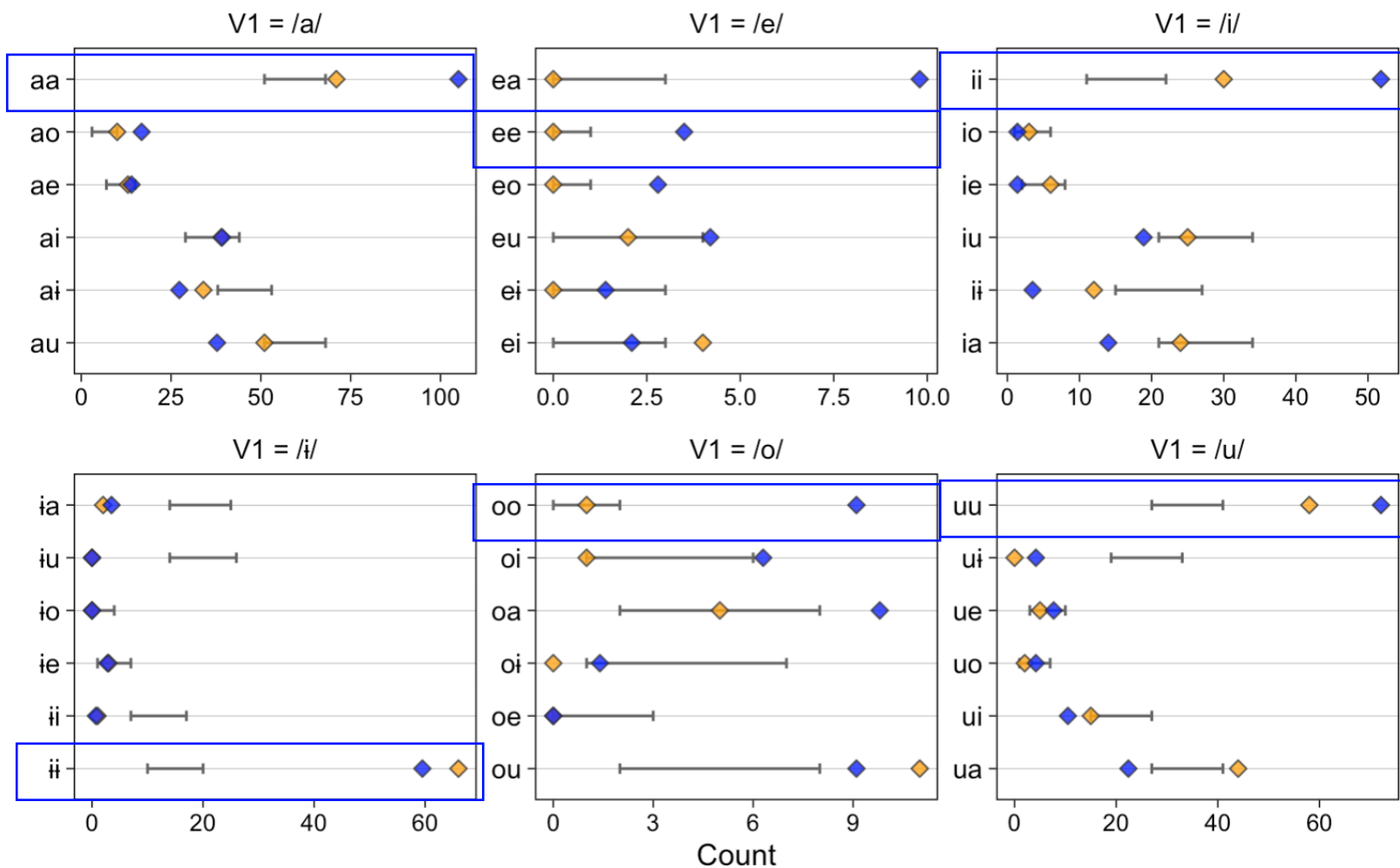
<i>UR</i>	<i>Old pattern</i>	<i>New pattern</i>
/sima/	sma: ~ ik-sima:	səma ~ ik-sima:
/rana/	rna: ~ ik-rana:	rəna ~ ik-rana:
/ɖamari/	ɖmari ~ ik-ɖamri:	ɖəmari ~ ik-ɖaməri:
/timusu/	tmusu ~ ik-timsu:	təmusu ~ ik-timəsu:
/giŋigiŋi/	gŋigŋi ~ ik-giŋgiŋi:	gəŋigəŋi ~ ik-giŋəgiŋi:

Monte Carlo simulations

Pooled over all vowels



Monte Carlo simulations



*CCC-avoidance

CC-

initial	prefix	type	n	p	example
no	i-	alternating	41	0.65	i-ubriṯ
		other	22	0.35	i-udaliṯ
no	ik-	alternating	68	0.68	ik-avṇiṯ
		other	32	0.32	ik-anuruṯ
yes	i-	alternating	163	0.79	i-kulbuṇuṯ
		other	44	0.21	i-ckickiṯ
yes	ik-	alternating	321	0.91	ik-silviṯ
		other	32	0.09	ik-bl avniṯ

Historical explanations for failure to undergo syncope are not enough

Proto-Rukai diphthongs and V?V sequences generally fail to undergo syncope in modern Rukai:

Proto-Rukai		Maga	(regular)
*daʔanə →	danɪ	(dnɪː)	‘house’
*payso →	pesu	(psuː)	‘money’
*bəʔəkə →	bɪkɪ	(bkɪː)	‘pig’

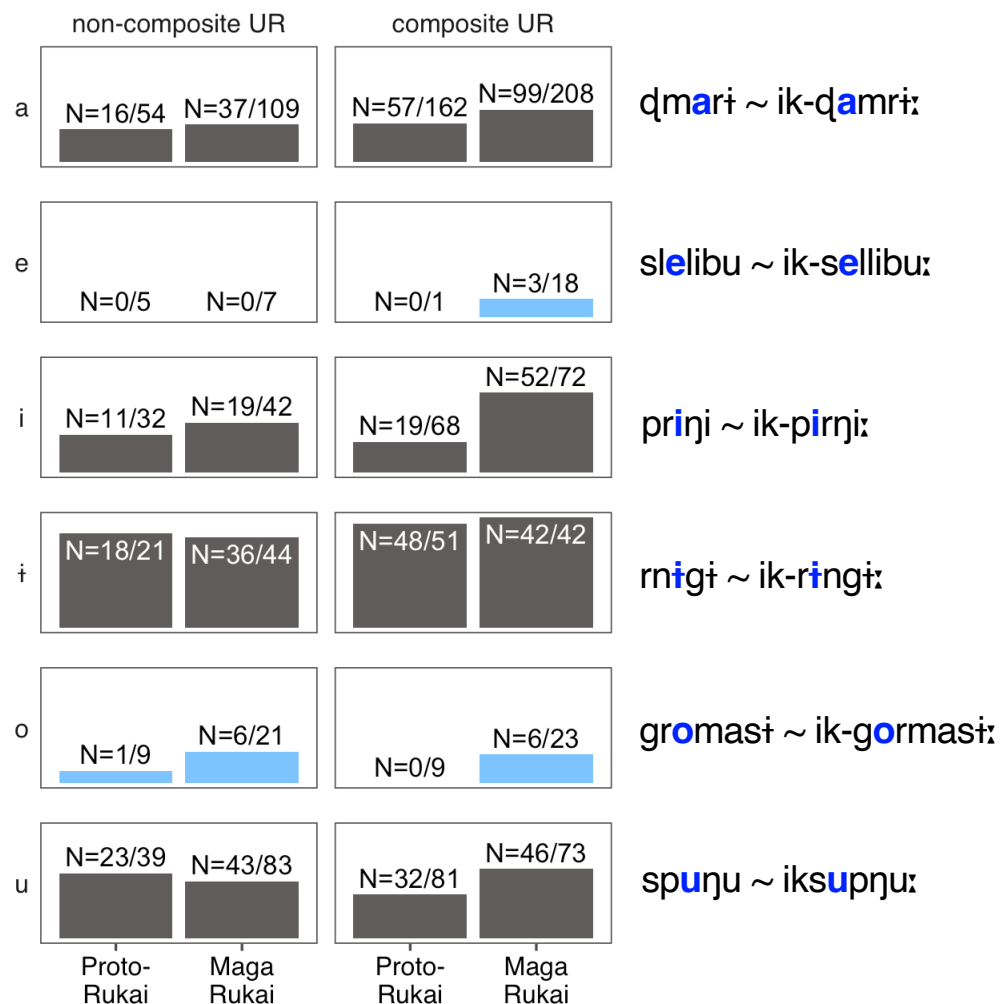
Only accounts for **35/70 (50%)** of stem allomorphs that have a known protoform and fail to undergo syncope.

*koroŋo →	kurŋu	(krŋu)	‘cloud’
*dokoʔo →	duku	(dkuː)	‘grow (plant)’
*pəkə →	pɪkɪ	(pkɪː)	‘house lizard’

Proportion of vowel-matching by vowel

True across **all vowel contexts** in composite URs

Even when **not historically matching**



Vowel-matching makes paradigms predictable from both surface allomorphs

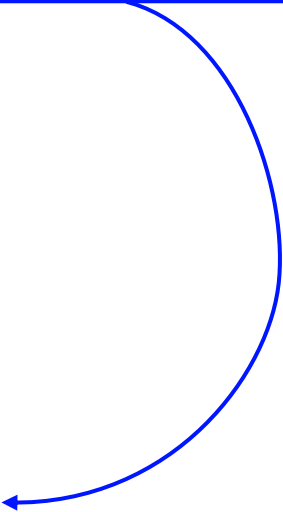
ORIGINAL UR TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	iri	i-iri:	?
stem-base	/abaki/	abaki	ik-abki:	?
negative-base	/rana/	rna:	ik-rana:	?
composite UR	/damari/	dmari	ik-damri:	?

What is the effect of vowel-matching on the surface predictability of these paradigms?

Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	
negative-base	/rana/	r a	ik-r a a:	
composite UR	/damari/	q a ri	ik-q a mri:	

No effect: paradigm was already
predictable from both allomorphs



Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	
negative-base	/rana/	r a	ik-r a a:	
composite UR	/damari/	d a ma i	ik-d a ma i :	

UR (=stem)	/abaki/	/ik- a baki-V/
Deletion	--	ik- a bØki:
SR	[abaki]	[ik- a bki:]

If **UR=stem**, the negative form can be derived via **deletion**

Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	
negative-base	/rana/	rna a	ik-r a na:	
composite UR	/damari/	dma a ri	ik-d a mri:	

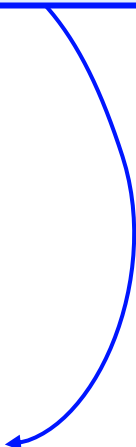
UR (=negative)	/a ₁ bki/	/ik-a ₁ bki-V/
Vowel copying	a ₁ ba ₁ ki	--
SR	[abaki]	[ik-abki:]

If **UR=negative**, the stem form can be derived via **copying**

Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	both allomorphs
negative-base	/rana/	rna:	ik-r a na:	
composite UR	/damari/	dma:ri	ik-d a mri:	

paradigm is now predictable
from both allomorphs



Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	both allomorphs
negative-base	/rana/	rna:	ik-r a na:	
composite UR	/damari/	dma:ri	ik-d a mri:	

UR (=negative)	/r a na/	/ik-r a na-V/
Deletion	rØn a	--
SR	[rn a]	[ik-r a na:]

If **UR=negative**, the stem form can be derived via **deletion**

Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	both allomorphs
negative-base	/rana/	rna:	ik-r a na:	
composite UR	/damari/	dma:ri	ik-d a mri:	

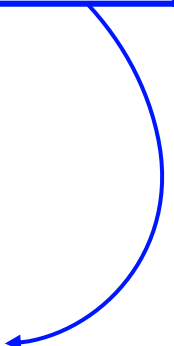
UR (=stem)	/rna/	/ik-rna ₁ -V/
Vow copying	--	ik-ra ₁ na ₁ :
SR	[rna]	[ik-r a na:]

If **UR=stem**, the negative form can be derived via **copying**

Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	both allomorphs
negative-base	/rana/	rna:	ik-r a na:	both allomorphs
composite UR	/damari/	dma:ri	ik-d a mri:	

Once again, the paradigm is now predictable from both allomorphs



Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	both allomorphs
negative-base	/rana/	rna:	ik-r a na:	both allomorphs
composite UR	/damari/	dma ₁ ri	ik-d a mri:	

UR (=stem)	/dm a ri/	/ik-dm ₁ a ₂ ri-V/
Metathesis	--	ik-d a ₂ m ₁ ri:
SR	[dm a ri]	[ik-d a mri:]

Both allomorphs can be derived from each other via **metathesis**

Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	both allomorphs
negative-base	/rana/	rna:	ik-r a na:	both allomorphs
composite UR	/damari/	dm a ri	ik-d a mri:	

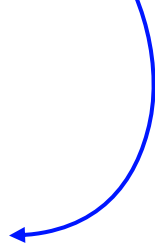
UR (=negative) /d**a**₁m₂ri/ /ik-d**a**mri-V/
Metathesis dm₂**a**₁ri --
SR [dm**a**ri] [ik-d**a**mri:]

Both allomorphs can be derived from each other via **metathesis**

Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	both allomorphs
negative-base	/rana/	rna:	ik-r a na:	both allomorphs
composite UR	/damari/	dma:ri	ik-d a mri:	both allomorphs

paradigm is now predictable
from both allomorphs



Vowel-matching makes some paradigms **predictable from both surface allomorphs**

TYPE	UR	STEM	NEGATIVE	PREDICTABILITY IF VOWEL-MATCHING
non-alternating	/iri/	i ri	i- i ri:	both allomorphs
stem-base	/abaki/	a baki	ik- a bki:	both allomorphs
negative-base	/rana/	rna:	ik-r a na:	both allomorphs
composite UR	/damari/	dma r i	ik-d a mrɪ:	both allomorphs

If vowel-matching is predictable, stem and negative allomorphs are ***predictable from each other, making URs concrete***